

Seeing Like Humans: Task-Driven Token Reduction for Efficient ViT in Robotic Navigation

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Abstract— In robotics, vision is critical for enabling agents to perceive and interact with the environment. Recent advancements in vision models, particularly Vision Transformers, have shown remarkable performance in pure vision tasks like object recognition and scene understanding, demonstrating great potential for robotic applications such as object navigation. However, their computational cost grows quadratically with respect to the number of tokens, posing significant challenges for real-time deployment on resource-constrained robotic platforms. To enhance ViT efficiency in robotic tasks, we propose a biologically inspired token reduction framework that dynamically allocates computation to task-relevant regions in images while neglecting those irrelevant regions. Our method introduces two key components: (1) a task-driven spatial attention mechanism that selectively prunes redundant tokens based on the current task, and (2) a temporal feature reusing module that reuses stable visual features across frames to minimize redundant computation. Together, these components enable the visual perception model to focus only on relevant regions, significantly improving inference speed. Experiments show that our method notably reduces inference time in object navigation and manipulation tasks without significant performance degradation. The proposed framework achieves efficient ViT inference on edge devices such as the NVIDIA Jetson Orin (Edge GPU) and Raspberry Pi (Embedded CPU), reaching 56.4 FPS and 1.95 FPS, respectively. This $1.5\times$ to $3\times$ speedup significantly advances the feasibility of real-time robotic vision on embedded platforms.

Index Terms—Deep Learning for Visual Perception, Perception-Action Coupling, Visual Learning.

I. INTRODUCTION

IN robotic systems operating in unstructured environments, real-time visual processing is critical for autonomous decision-making. Early robotic vision systems based on Convolutional Neural Networks (CNNs) were effective at extracting local features. However, the receptive fields of CNNs are inherently bounded by their fixed kernel sizes and remain static, making them struggle to adapt to variable-sized contextual dependencies based on visual content.

In contrast, Vision Transformers (ViTs) eliminate this limitation through content-dependent self-attention, which dy-

namically adjusts attention distribution on the fly to flexibly capture arbitrary spatial dependencies, providing a more holistic scene understanding. However, the quadratic complexity of self-attention leads to a severe trade-off: increasing input resolution improves visual understanding, but also raises inference latency. This in turn reduces control frequency, causing “command starvation” and an idle waiting delay during continuous deployment that may affect safety and locomotion stability. This challenge highlights the need for more efficient architectures that retain the benefits of ViTs while meeting the real-time requirements of robotics.

Biological vision efficiently processes complex scenes under strict metabolic constraints [1], [2]. Unlike standard artificial attention mechanisms that universally retain all tokens via dense $O(N^2)$ soft-weighting, biological vision selectively suppresses or deprioritizes redundant sensory inputs to minimize metabolic costs. Motivated by this, we propose a token pruning strategy that explicitly curtails computational complexity by physically pruning redundant tokens across space and time, guided by two biological principles: (1) **Predictive Coding** (Temporal Focus): Driven by *neural adaptation* [3], the visual cortex suppresses neural firing for unchanging stimuli—much like humans effortlessly ignore a static background but instantly react to a moving obstacle. Similarly, robotic perception should recycle stable features across sequential frames, restricting dense computation to novel changes. (2) **Top-Down Modulation** (Spatial Focus): Governed by the prefrontal cortex [4], human vision actively suppresses task-irrelevant pathways rather than relying on passive bottom-up saliency. This manifests as *inattention blindness* [5], where humans focusing on a specific target naturally filter out irrelevant clutter. Analogously, explicit computational resources in ViTs should be actively directed to regions particularly critical for the robotic intent.

Building on these axioms, we propose a bio-inspired token reduction method. It employs temporal token reuse to minimize inter-frame redundancy and task-driven pruning to eliminate intra-frame irrelevant regions. (1) The temporal token reusing module identifies persistent features across frames, leveraging cross-attention to propagate information and minimize redundant computation. (2) The task-driven token pruning module evaluates the relevance of each token based on the long-term goals and short-term operational phases. By focusing on task-relevant regions, such as target objects or navigable paths, it prunes unnecessary tokens.

Unlike prior methods that rely on task-agnostic metrics (e.g., purely bottom-up visual saliency or high-frequency edge preservation) [6], [7] and static objectives, our approach per-

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forms token-level pruning guided by temporal redundancy and task-specific requirements.

Evaluated on object navigation and precision manipulation tasks, our reduction strategy achieves a $1.5\text{--}3\times$ speedup across standard edge platforms. Crucially, in real-world deployment on a quadruped robot, our method successfully reduces stagnant waiting periods between actions, thereby enabling near-seamless transitions and faster task completion as shown in the supplementary video. In summary, our main contributions are as follows:

- 1) We proposed a task-driven token pruning module that physically drops tokens based on top-down intent, explicitly reallocating compute to task-relevant tokens and filtering out redundant ones.
- 2) We introduced a temporal token reusing module that recycles stable visual features across frames, reducing redundant extraction during video processing.
- 3) We demonstrated the effectiveness of our method through extensive experiments, showing significant real-time performance across multiple platforms and robotic tasks, effectively preventing control stuttering and accelerating total physical execution time.

II. RELATED WORK

A. Visual Perception in Robotics

Vision Transformers (ViTs) [8] excel at modeling global spatial dependencies for complex embodied tasks like manipulation and navigation [9], [10]. However, deploying these dense architectures on resource-constrained edge platforms introduces a severe bottleneck unique to closed-loop robotics: the quadratic computational complexity of self-attention directly couples perception latency with physical execution. High inference overhead lowers the achievable control loop frequency, inducing command starvation and stagnant idle delays after each execution step. This delayed feedback loop compromises real-world safety and locomotion stability, creating a fundamental tension between perceptual fidelity and real-time control responsiveness. Consequently, mitigating this heavy computational burden through efficient architecture design or model acceleration is highly necessary to sustain real-time execution in edge robotics.

B. ViT Acceleration Methods

General ViT acceleration involves architectural optimizations like model pruning [11], quantization [12], and efficient attention (e.g., window-based [13] or linear [14]). While these reduce the static computational footprint, they remain agnostic to the input data's dynamic content.

Token reduction dynamically addresses the quadratic complexity of ViTs by reducing sequence length during inference. Existing strategies primarily fall into informativeness-based pruning and similarity-based merging. To identify uninformative tokens, prediction-based methods evaluate intermediate features (e.g., SPViT [15], AdaViT [16]), while attention-based methods utilize attention weights (e.g., ATS [17], EViT [6]). These less critical tokens are then discarded entirely or routed via skip connections (e.g., DoViT [18]).

Conversely, similarity-based methods assume highly correlated tokens contain redundant information. Techniques like ToME [19], ALGM [20], ATM [7], and LGTM [21] group these tokens using clustering, adaptive thresholds, or frequency-aware merging strategies to preserve overall spatial integrity while reducing token counts.

Despite their efficiency on static computer vision benchmarks, existing methods lack the task-driven awareness and temporal adaptability required for embodied robotics. First, saliency-based token pruning [6], [16] often discards low-saliency but task-critical details, causing failures in closed-loop control. Second, bottom-up similarity or frequency-aware token merging methods [7], [21] compute metrics entirely agnostic to the agent's current state and task, wasting precious onboard resources on compressing irrelevant backgrounds or preserving high-frequency textures (e.g., floor patterns) that offer no actionable navigation value.

Furthermore, beyond these spatial and intent-agnostic limitations, existing token reduction frameworks suffer from a critical temporal inefficiency when applied to continuous robotic streams. Because these methods process each incoming frame in isolation as a static image, they completely overlook the temporal redundancy across sequential observations, forcing the onboard processor to redundantly re-compute stable visual features from scratch. In contrast, our framework comprehensively addresses these limitations across both temporal and spatial dimensions. Temporally, our token reusing module is explicitly tailored for streaming, online deployment, operating as a causal pipeline that actively leverages past computational results to accelerate immediate next-frame extraction. Spatially, our top-down modulation modulates token survival based on the current state $\mathbf{h}_{t-1}$ and global active intent $\mathbf{t}$, successfully bridging the gap between passive, isolated perception and active, causal robotic control.

III. METHOD

As discussed in section II-B, existing approaches often inadvertently prune task-critical information when relevant content lacks visual prominence, while retaining irrelevant but salient regions. This fundamental misalignment results in either compromised success rates or significant computational inefficiencies during real-world deployment.

In contrast, our method introduces task-driven token pruning, which dynamically adjusts to the specific needs of the current situation. This approach ensures that only the most relevant tokens are retained, improving processing efficiency without sacrificing success rates. As shown in Figure 1, our method employs two pruning modules to minimize redundant computation while preserving task-critical features. By selectively retaining tokens based on task relevance and temporal coherence, our system ensures efficient feature extraction, making it suitable for real-world robotic applications.

A. Temporal Token Reusing

In robotic environments, consecutive frames are often similar, with only small changes in the environment. Reprocessing redundant information can be computationally expensive. To

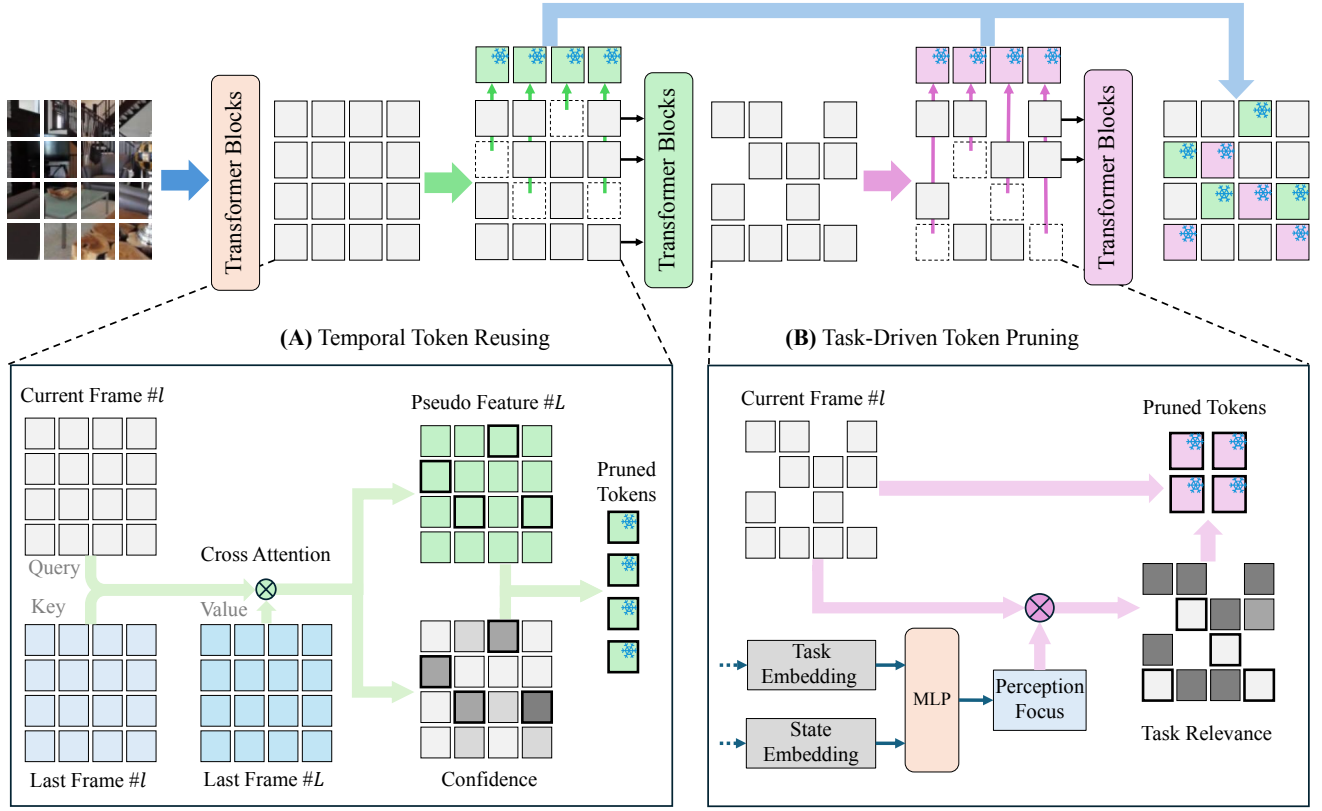


Fig. 1: Overview of the proposed method. Given an input frame, the process consists of two pruning stages: (A) Temporal Token Reusing, where several tokens are identified as reusable based on the feature map of the previous frame. These tokens directly inherit the final features from the previous frame’s extraction result, bypassing further processing. (B) Task-Driven Token Pruning, where tokens irrelevant to the task at hand are filtered out based on the short-term temporal state $\mathbf{h}_{t-1}$ and the long-term target object $\mathbf{t}$. These tokens are also excluded from subsequent feature extraction.

address this, we propose a *temporal token reusing module*, which leverages feature consistency between adjacent frames. By reusing the final features from the previous frame, we approximate the current frame’s output, avoiding unnecessary computation of redundant tokens. This approach can efficiently reduce computational overhead by focusing only on newly observed elements.

Let $\mathbf{F}^{(t)} = \{\mathbf{f}_i^{(t)}\}_{i=1}^N$ denote the token features at a given Transformer layer for the current frame t , and $\mathbf{F}^{(t-1)} = \{\mathbf{f}_j^{(t-1)}\}_{j=1}^N$ the corresponding features for the previous frame. Let $\mathbf{Z}^{(t-1)} = \{\mathbf{z}_j^{(t-1)}\}_{j=1}^N$ represent the final output features of the previous frame.

We treat the current frame’s tokens $\mathbf{F}^{(t)}$ as *queries*, the previous frame’s features $\mathbf{F}^{(t-1)}$ as *keys*, and its final features $\mathbf{Z}^{(t-1)}$ as *values* in a cross-attention operation. The propagated feature for the i -th token in frame t is computed as:

$$\hat{\mathbf{z}}_i^{(t)} = \text{CrossAttn}(\mathbf{f}_i^{(t)}, \mathbf{F}^{(t-1)}, \mathbf{Z}^{(t-1)}). \quad (1)$$

In this mechanism, each token in the current frame searches for similar tokens in the previous frame, using the attention scores between the queries (current frame tokens) and the keys (previous frame tokens) to determine which features from the previous frame are most relevant. The final feature for the i -th token in the current frame is then predicted based

solely on the most relevant features from the previous frame, effectively bypassing the need for further computation of redundant tokens.

To determine whether a token $\mathbf{f}_i^{(t)}$ has a reliable match in the previous frame, we compute a confidence score r_i based on three factors: (1) The entropy of attention weights $\mathbf{a}_i$, measuring the sharpness of the match. (2) The cosine similarity between predicted $\hat{\mathbf{z}}_i^{(t)}$ and the current-layer feature $\mathbf{f}_i^{(t)}$. (3) The maximum attention weight $\max(\mathbf{a}_i)$, indicating the strength of the most confident correspondence.

These values are passed through a small MLP predictor $Q(\cdot)$ to produce a confidence score:

$$r_i = Q\left(\left[H(\mathbf{a}_i), \cos\left(\hat{\mathbf{z}}_i^{(t)}, \mathbf{f}_i^{(t)}\right), \max(\mathbf{a}_i)\right]\right), \quad r_i \in [0, 1]. \quad (2)$$

We then use a predefined threshold thr_{temp} to identify reusable tokens:

$$\mathcal{I}_{\text{reuse}} = \{i \mid r_i \geq \text{thr}_{\text{temp}}\}, \quad (3)$$

$$\mathcal{I}_{\text{process}} = \{i \mid r_i < \text{thr}_{\text{temp}}\}. \quad (4)$$

At inference time, tokens in $\mathcal{I}_{\text{reuse}}$ are excluded from further Transformer computation. Their propagated features $\hat{\mathbf{z}}_i^{(t)}$ are

directly passed to the final feature map. To preserve their influence, we enhance the remaining tokens by replacing $\mathbf{f}_i^{(t)}$ with $\hat{\mathbf{z}}_i^{(t)}$. This allows the retained tokens to integrate information from those that were confidently pruned, effectively transferring information extracted in previous frames.

B. Task-Driven Token Pruning

In robotic tasks, the relevance of visual information varies depending on the current stage of the task. For instance, during the exploration phase of navigation, the robot needs to focus on detecting open spaces, obstacles, and potential pathways, while ignoring static, irrelevant background elements like walls. In contrast, when approaching the target, the robot must dynamically prioritize features related to the current task—such as target objects or mission-specific landmarks—and minimize processing of peripheral elements as task requirements evolve.

To achieve this, we introduce a *task-driven token pruning module* that adaptively removes irrelevant tokens based on the hierarchical alignment of long-term task objectives and the current short-term execution phase. By integrating a lightweight gating mechanism into the Transformer architecture, the model is able to selectively predict and retain task-relevant features. This ensures maximum computational efficiency by focusing on the most critical information at each consecutive stage of the task.

Let the input token features at a certain Transformer layer be denoted as $\{\mathbf{f}_i\}_{i=1}^N$, where $\mathbf{f}_i \in \mathbb{R}^d$ is the feature of the i -th token, and N is the total number of tokens. To estimate the dynamic perception focus vector $\mathbf{c} \in \mathbb{R}^d$, we employ a lightweight predictor $P(\cdot)$ conditioned on both the task target representation $\mathbf{t}$, encoded as a one-hot vector and the previous hidden state of the downstream recurrent neural network $\mathbf{h}_{t-1} \in \mathbb{R}^{d_h}$. Crucially, while $\mathbf{t}$ establishes the overarching long-term global target, $\mathbf{h}_{t-1}$ implicitly captures the current multi-stage navigation phase to infer immediate, short-term subgoals. Since long-horizon navigation naturally consists of distinct tactical phases (e.g., room exploration vs. local target approaching) that demand entirely different visual attention priorities, $\mathbf{h}_{t-1}$ provides the vital temporal context to adaptively shift the focus step-by-step. The predictor $P(\cdot)$, implemented as an MLP, fuses these hierarchical semantic and temporal inputs to project them into the high-dimensional visual feature space:

$$\mathbf{c} = P(\mathbf{h}_{t-1}, \mathbf{t}) = \text{MLP}([\mathbf{W}_h \mathbf{h}_{t-1}, \mathbf{W}_t \mathbf{t}]), \quad (5)$$

where $\mathbf{W}_h$ and $\mathbf{W}_t$ are learnable linear projection matrices, and $[\cdot]$ denotes the concatenation operation. In this formulation, $\mathbf{c}$ encodes the semantic characteristics that the model should attend to under the current situation.

For each token $\mathbf{f}_i$, we quantify its task-relevance score α_i by computing the cosine similarity between the token and the focus $\mathbf{c}$:

$$\alpha_i = \cos(\mathbf{f}_i, \mathbf{c}) = \frac{\mathbf{f}_i^\top \mathbf{c}}{\|\mathbf{f}_i\| \cdot \|\mathbf{c}\|}. \quad (6)$$

Tokens are then partitioned into a set of retained indices $\mathcal{I}_{\text{keep}}$ and pruned indices $\mathcal{I}_{\text{drop}}$. To consistently couple smaller

thresholds with more aggressive pruning across both modules, we formulate the pruning condition using a dynamic similarity cutoff $1 - \text{thr}_{\text{task}}$:

$$\mathcal{I}_{\text{keep}} = \{i \mid \alpha_i \geq 1 - \text{thr}_{\text{task}}\}, \quad (7)$$

$$\mathcal{I}_{\text{drop}} = \{i \mid \alpha_i < 1 - \text{thr}_{\text{task}}\}. \quad (8)$$

Under this formulation, tokens exceeding the cutoff are actively retained for downstream Transformer blocks, while those falling below are considered irrelevant and therefore directly bypassed.

At inference time, only the relevant tokens in $\mathcal{I}_{\text{keep}}$ contribute to the computation of subsequent Transformer blocks. The irrelevant tokens in $\mathcal{I}_{\text{drop}}$ are excluded from further processing; their features are directly passed on to the final feature map, bypassing any additional Transformer layers. This results in faster inference while maintaining the integrity of task-relevant computations.

C. Implementation Details

Sequential Decision Architecture: To handle temporal dynamics and generate sequential actions, we append a recurrent neural network (RNN) after the Transformer backbone. Specifically, the retained output tokens from the ViT are spatially rearranged back into a $C \times H \times W$ feature map. This map is then processed by a 2-layer downsampling Convolutional Neural Network (CNN) to reduce spatial dimensions. The resulting condensed features are then flattened and concatenated with auxiliary sensor features, including GPS coordinates, compass headings, and the target object embeddings before being fed into the RNN, forming an end-to-end ViT+RNN policy.

Temporal Token Reusing: This module consists of a cross-attention block and a 2-layer MLP Q for predicting the confidence score r_i of temporal matches. It is pretrained using video sequences from the simulator. The training loss combines a regression term and a classification term:

$$\mathcal{L}_{\text{reuse}} = \|\hat{\mathbf{z}}_i - \mathbf{z}_i\|_2 + \text{CE} \left(r_i, 1 - \text{sat} \left(\frac{\|\hat{\mathbf{z}}_i - \mathbf{z}_i\|_2}{M} \right) \right), \quad (9)$$

where the first term is a standard regression loss between the predicted feature $\hat{\mathbf{z}}_i$ and the ground-truth ViT feature $\mathbf{z}_i$. The second term supervises the confidence score r_i . To generate a target for this confidence, we transform the regression error into a pseudo-label by normalizing it with a dynamic threshold M , where $\text{sat}(x) = \min(1, \max(0, x))$ denotes the saturation function. Based on our empirical observation of a sharp knee point in the sorted MSE distribution, we set M to the value at this point, which is located using a knee-detection algorithm. This normalized error serves as a soft label to train the confidence prediction part.

Task-Driven Token Pruning: We train the focus predictor P (a 2-layer MLP) using task-level supervision, where the agent is optimized to complete a downstream objective. The pruning decisions are guided by gradients derived from task-specific losses, allowing the model to learn which tokens are essential for the current situation. This setup supports both reinforcement learning and imitation learning paradigms. In

our implementation, we follow the settings of prior works [22] and adopt imitation learning using demonstration trajectories collected in simulation [23].

To enable end-to-end optimization, we adopt the Straight-Through Estimator [24] to handle the non-differentiable token selection step. The binary retention mask $\tilde{m} = [\tilde{m}_1, \dots, \tilde{m}_N]$ is generated based on the calculated similarity α_i , and gradients are propagated through the sampling operation by treating it as the identity function during backpropagation.

In addition to the task loss, we introduce a sparsity loss to reduce the number of retained tokens. Specifically, we apply an ℓ_1 penalty on the retention mask:

$$\mathcal{L}_{\text{sparse}} = \lambda \cdot \frac{1}{N} \sum_{i=1}^N \tilde{m}_i, \quad (10)$$

where λ controls the trade-off between accuracy and efficiency. The final training objective is the sum of the task loss and the sparsity loss:

$$\mathcal{L}_{\text{prune}} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{sparse}}. \quad (11)$$

This formulation allows the model to balance task performance with constrained computational efficiency in a fully differentiable manner.

IV. EXPERIMENTS

A. Experimental Setup

We evaluate our token reduction method on the ObjectNav task from Habitat Challenge 2022, which involves navigating to six object categories (bed, chair, toilet, TV, sofa, plant) across 88 Gibson scenes, with 2,000 validation episodes. Imitation learning is performed using 77k human demonstration trajectories collected via Habitat-Web [23].

Following prior works [22], [25], we adopt an imitation learning paradigm to train the policy. The visual backbone of our model is a ViT, specifically the DeiT-Tiny [26] model pretrained on ImageNet. To handle temporal dynamics and generate sequential actions, we append a recurrent neural network (RNN) after the Transformer backbone. Specifically, the retained output tokens from the ViT are spatially rearranged back into a $C \times H \times W$ feature map. This map is then processed by a 2-layer downsampling Convolutional Neural Network (CNN) to reduce spatial dimensions. The resulting condensed features are flattened and subsequently fed into the RNN, forming an end-to-end ViT+RNN policy. We insert the temporal token reusing module after the 3rd layer of the ViT and the task-driven token pruning module after the 6th layer, both with thresholds of 0.5 and a sparsity weight $\lambda = 0.1$.

We compare our method against the following baselines:

- **Baseline**: A ViT+RNN model without pruning.
- **Baseline†**: A smaller variant of the baseline with comparable computational complexity to our pruned model.
- **EViT** [6]: A classic token pruning method that drops low-attention tokens based on their attention to the [CLS] token.
- **ToMe** [19]: A token merging approach that fuses the most similar token pairs to reduce the number of tokens.

- **DynamicViT** [27]: A token pruning method that employs a learnable network to assign importance scores to tokens and subsequently discards those with low scores.
- **ATM** [7]: A recent training-free Adaptive Token Merging method that dynamically adjusts layer-specific similarity thresholds to minimize information loss.
- **LGTM** [21]: A frequency-aware token compression strategy that preserves high-frequency details by discouraging the merging of high-contrast tokens.

All trainable methods are trained under identical conditions to ensure a fair comparison, while training-free methods use their original inference procedures.

We report four metrics to evaluate both navigation performance and computational efficiency:

- **Frames Per Second (FPS)**: measuring inference speed.
- **GFLOPs**: measuring the amount of computation required for model inference. It indicates how many floating-point calculations are performed.
- **Success Rate (SR)**: the percentage of episodes where the agent reaches the target object successfully within the time budget.
- **Success weighted by Path Length (SPL)**: a metric that accounts for both success and path optimality. The shorter the path, the higher the SPL.

B. Main Results

As shown in Table I, our method consistently outperforms existing approaches in both efficiency and task performance. On the NVIDIA Orin platform, our full model achieves 56.4 FPS, representing a 50.8% improvement over the baseline and clearly surpassing other approaches. Notably, while recent advanced merging methods like ATM and LGTM preserve competitive success rates, their complex inference-time calculations, such as adaptive threshold optimization across layers or Laplacian-gated frequency separation, introduce significant computational overhead. This bottlenecks their actual deployment speed to approximately 19 FPS on edge GPUs. The advantage becomes even more pronounced on the resource-constrained Raspberry Pi platform, where our method reaches 1.95 FPS, nearly tripling the baseline speed (2.83×). This scalability highlights the practicality of our approach for real-world robotic systems with limited computational resources.

In terms of computational cost, our method achieves the lowest complexity with 5.00 GFLOPs. Compared to alternative pruning-based approaches that remain above 6 GFLOPs and the baseline's 8.99 GFLOPs, this reduction translates directly into lower energy consumption and faster inference, reinforcing the efficiency of our design.

Equally important, our method preserves navigation effectiveness. While several baselines exhibit severe degradation in Success Rate (SR) and Success weighted by Path Length (SPL), our method maintains 50.05% SR and 21.46% SPL. This corresponds to only a 2.05% and 0.57% absolute drop from the baseline, respectively—substantially smaller than the large declines observed in prior methods. Such results confirm that our acceleration does not come at the expense of reliable navigation behavior.

TABLE I: Performance comparison of different methods on object navigation tasks, evaluated on both Orin (GPU) and Raspberry Pi (CPU) platforms. (sp) denotes the variant using only the Task-Driven Pruning, and (temp) denotes the variant using only the Temporal Reusing.

	FPS		GFLOPs	SPL(%)	SR(%)
	Orin (GPU)	Pi (CPU)			
Baseline	37.4	0.69	8.99	22.03	52.10
Baseline [†]	48.6	1.17	6.42	13.76	35.25
EViT [6]	47.7	1.04	6.88	15.42	41.80
ToMe [19]	40.9	1.20	6.02	21.29	50.10
DynamicViT [27]	47.9	1.05	6.89	18.08	46.40
ATM [7]	19.2	1.48	6.13	21.10	50.60
LGTM [21]	19.5	1.13	7.01	21.35	50.90
Ours(sp)	48.1	1.35	6.86	21.05	51.50
Ours(temp)	47.2	1.16	6.87	21.54	51.20
Ours(full)	56.4	1.95	5.00	21.46	50.05

Overall, the results demonstrate that our method achieves a highly adaptive acceleration that scales across diverse hardware. Unlike static or task-agnostic baselines, our design dynamically reconfigures computational effort by exploiting spatio-temporal redundancy. This ensures that perceptual resources are strictly reserved for task-critical information, thereby maintaining navigation reliability even under the stringent constraints of real-world robotic deployment.

C. Evaluation on Manipulation Task

To verify generalization beyond coarse-grained navigation, we evaluated on the `Reach-v3` task from MetaWorld (single-task MT1 setting, no task identifier needed). We first trained a state-based SAC expert on MetaWorld’s native low-dimensional observations and collected deterministic successful rollouts from the `corner2` camera, recording RGB images and continuous 4-D actions. All vision-based agents were then trained by imitation learning on these trajectories, with the same architecture and preprocessing as in Section IV-A except for a linear head outputting 4-D continuous actions; training used 64-step sequences, batch size 8, Smooth L1 regression with additional weighting on action-inflection steps, and AdamW, with periodic in-environment evaluation. As shown in Table II, standard token pruning methods suffer particularly dramatic drops: DynamicViT collapses to 38.0% success, as task-agnostic saliency estimation discards vital spatial features of the end-effector or target object, while ATM and LGTM retain moderate success ($\sim 62\%$) but at only 19 FPS, unsuited for high-frequency control. In contrast, our task-driven framework maintains a competitive 66.0% success rate at 50.81 FPS on the Jetson Orin, a 40% speedup over the baseline, confirming that our approach generalizes across fundamentally different embodiments.

D. Quantitative Validation of Top-Down Modulation

To quantitatively validate the bio-inspired motivation, specifically that our mechanism mimics human top-down attentional modulation rather than standard bottom-up saliency, we analyzed token retention distributions under varying intentions within the exact same visual input.

Table III demonstrates the percentage of tokens retained for specific semantic regions across 1,000 validation frames when

TABLE II: Performance comparison on the MetaWorld Reach-v3 manipulation task.

	FPS		SR(%)
	Orin (GPU)	Pi (CPU)	
Baseline	36.26	0.71	68.0
ToMe [19]	40.49	1.39	61.0
DynamicViT [27]	35.72	1.58	38.0
ATM [7]	19.03	1.41	62.0
LGTM [21]	18.95	1.44	63.0
Ours(full)	50.81	1.76	66.0

TABLE III: Percentage of tokens retained within each semantic region, normalized by the total number of tokens belonging to that region, under different task goals.

Current Goal	Chair Tokens	Bed Tokens	Others
Find Chair	72%	24%	12%
Find Bed	13%	70%	14%

the input command shifts. When the instruction is “Find Bed”, the network heavily focuses computational resources on the bed region, yielding a token retention rate of 70%. Crucially, when the goal shifts to “Chair” for the identical image, the retention of bed tokens drops precipitously to 24%, while chair tokens spike to 72%. Standard attention mechanisms perform dense $O(N^2)$ computation universally, whereas our task-driven predictor physically suppresses computation associated with up to 76% of irrelevant background tokens per frame. This provides measurable validation that our method successfully instantiates a top-down, active computational resource allocation strategy that strictly aligns with human cognitive expectations of goal-directed attention.

E. Real-World Robot Deployment

To validate practical real-world deployment, we deployed the policy on a Unitree Go2 quadruped equipped with an onboard NVIDIA Orin NX platform. The system operates in a synchronous sequential control loop, where each discrete action primitive (0.25 m forward step or 30° turn) blocks the next command until a fresh visual inference completes. As quantified in Table IV, the unpruned baseline requires 451 ms of inference overhead, severely starving the command queue and forcing the robot into a jarring stop-and-go motion. In contrast, our task-driven pruning slashes onboard latency to

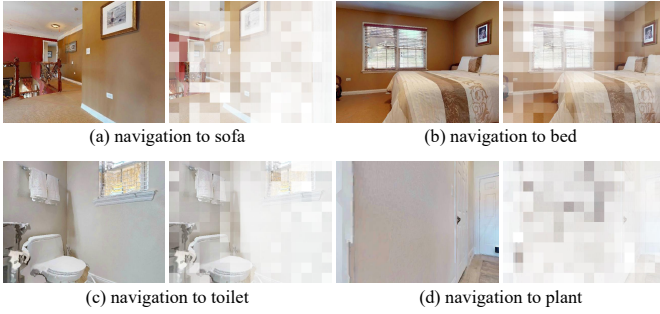


Fig. 2: Visualization of token pruning. Each pair shows the original image on the left and the corresponding probability map on the right. More transparent regions indicate a higher probability of token pruning.

172 ms, effectively matching the physical execution cycle of the quadrupedal gait. This optimization reduces stagnant idle delays, enabling near-seamless step transitions and minimizing total task completion time as demonstrated in the supplementary video.

TABLE IV: Onboard inference time (ms) comparison across different token reduction methods on the Unitree Go2.

Baseline	ToMe	DynamicViT	ATM	LGTM	Ours
451	343	365	283	327	172

F. Token Pruning Visualization

To better illustrate the underlying mechanism, we visualize the pruning probability maps in Figure 2. The results show that the model consistently focuses on task-relevant regions (e.g., target objects, obstacles) while discarding uninformative background areas. This task-aware selection effectively reduces spatial redundancy and retains critical visual cues, which explains why our method can substantially increase inference speed without degrading navigation success rate.

G. Ablation Study

Module Ablation. The experimental results in Table I demonstrate how each component contributes to the overall performance. Using only the task-driven token pruning (Ours (sp)) increases FPS to 48.1 (+28.6%) with minimal drop in success rate (51.5%), while temporal token reuse (Ours (temp)) achieves 47.2 FPS with slightly higher SPL than Ours (sp) (21.54% vs. 21.05%), highlighting its benefit in preserving features across frames. The full model combines both, reaching 56.4 FPS (+51%) and maintaining success above 50%, confirming that both components are crucial for optimal performance.

Thresholds Ablation. Table V further investigates the effect of key thresholds. The performance remains relatively stable across different temporal (thr_{temp}) and task-related (thr_{task}) thresholds, demonstrating the robustness of our approach. These results are obtained by directly adjusting the thresholds on the trained model without retraining, which also provides

TABLE V: Effect of varying temporal (top) and task (bottom) thresholds.

thr_{temp}	0.3	0.4	0.5	0.7	0.9
SR (%)	49.0	50.0	50.1	51.1	51.8
FPS	57.1	56.7	56.4	49.3	41.6

thr_{task}	0.3	0.4	0.5	0.7	0.9
SR (%)	42.8	50.1	50.1	50.3	51.4
FPS	57.3	57.0	56.4	49.7	38.7

TABLE VI: Ablation study of different confidence cues for temporal token reuse and their impact on reconstruction fidelity and computational efficiency.

Confidence Cues	Reuse MSE ↓	Reuse Ratio ↑
All cues (ours)	0.177	39%
w/o entropy	0.251	40%
w/o cosine similarity	0.179	35%
w/o max attention	0.262	39%

a simple and effective way to control the trade-off between success rate and speed.

Token Reuse Criteria Ablation. To quantify the contribution of each input cue in the temporal token reuse decision, we conduct an ablation study. The three cues considered are: (1) attention entropy, (2) maximum attention, and (3) cosine similarity between reconstructed and original token features.

We evaluate two metrics: (1) Reuse MSE, the mean squared error between reused token features and ground-truth features, indicating reuse accuracy; and (2) Reuse Ratio, the average percentage of tokens reused, indicating potential computational speed-up. Table VI reports the results. Removing any single cue increases either the reuse error or reduces the reuse ratio, confirming that all three cues are necessary for accurate and stable token reuse.

Robustness and Architecture Choices. Beyond static metrics, we evaluated the framework’s stability under non-ideal conditions. As compiled in Table VIII, under simulated dynamic perturbations like frame-skipping and Gaussian noise ($\sigma = 0.1$), the temporal module adaptively lowers its token reuse rate from 39% to 33% to maintain feature fidelity, demonstrating strong resilience against sensor degradation. Furthermore, architectural ablations on module placement (Table VII) confirm that our default configuration of inserting reuse at Layer 3 and pruning at Layer 6 achieves the optimal trade-off between computational throughput and task performance. Moving temporal reuse deeper to Layer 6 yields diminishing computational returns by dropping the inference speed to 54.6 FPS, while applying task-driven pruning prematurely at Layer 3 disrupts high-level semantic representations, causing the navigation success rate to dramatically collapse to 35.10%.

V. CONCLUSION

In this work, we introduced a biologically inspired token reduction framework to optimize the efficiency of Vision Transformers for real-time robotic applications, achieving a $1.5\times$ to $3\times$ edge acceleration without sacrificing essential

TABLE VII: Ablation study on the insertion layers of Temporal Reuse and Task-Driven Pruning modules.

Reuse Layer	Prune Layer	FPS (Orin)	Success Rate (%)
L3	L3	57.3	35.10
L6	L6	54.6	50.20
L3 (Default)	L6 (Default)	56.4	50.05

TABLE VIII: Robustness of the temporal token reuse module under dynamic perturbations.

Condition	MSE ↓ (Error)	Reuse Rate ↑ (Speed)
Normal	0.177	39%
Frame-skipping	0.184	38%
Gaussian noise	0.189	33%

performance across both navigation and manipulation tasks. Crucially, real-world quadrupedal deployment demonstrates that our method slashes single-step decision latency from 0.4 s to 0.2 s, effectively eliminating stagnant idle delays. Moving forward, extending this task-driven token management paradigm to larger Vision-Language-Action (VLA) architectures represents a promising direction, potentially generalizing active visual selection to multi-modal foundational intelligence for resource-constrained embodied agents.

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